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TASK 1 :

CampusEats integrates with an external payment partner named **PaySecure** to handle payment transactions. The single operation being integrated is **charge**, which is used to deduct the order amount from the student's payment method.

This edge is implemented using SOAP instead of REST because payment processing requires a strong and rigid contract, message-level security (credentials are passed in the SOAP Header), and reliable transactional behavior. Unlike other parts of CampusEats that use REST for flexibility and simplicity, the payment gateway needs guaranteed message structure, security, and clear success/failure semantics, which SOAP provides better through its formal WSDL contract and built-in fault handling.

Task 4 :

HTTP Request Block:

POST /soap/v1 HTTP/1.1

Host: api.paysecure.example.com

Content-Type: text/xml; charset=utf-8

SOAPAction: "http://paysecure.example.com/ws/charge"

Content-Length: 612

<?xml version="1.0" encoding="UTF-8"?>

<soap:Envelope

xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/"

xmlns:tns="http://paysecure.example.com/ws">

<soap:Header>

<tns:AuthHeader>

<Username>campuseats_user</Username>

<Password>securePass@123</Password>

</tns:AuthHeader>

</soap:Header>

<soap:Body>

<tns:chargeRequest>

<orderId>ORD-78945</orderId>

<amount>450.00</amount>

<currency>INR</currency>

<cardNumber>4111111111111111</cardNumber>

<expiryDate>12/28</expiryDate>

<cvv>123</cvv>

</tns:chargeRequest>

</soap:Body>

</soap:Envelope>

Task 5 :

1. Discovery Explanation

Discovery

CampusEats can obtain the PaySecure WSDL in two modern ways:

1. Direct URL

The WSDL is published at a well-known location:

<https://api.paysecure.example.com/wsd/PaySecureService.wsdl>

2. Catalogue / Registry Lookup

CampusEats looks up the service in an internal service catalogue (or API registry).

The catalogue returns the business details, service name, endpoint, and the WSDL location.

In modern systems, we no longer use a full UDDI server. Instead, a simple service catalogue or API registry is used.

2. Registry Entry (tModel style)

Business Name : PaySecure Gateway Pvt. Ltd.
Service Name : PaySecureService
Endpoint URL : <https://api.paysecure.example.com/soap/v1>
WSDL Location : <https://api.paysecure.example.com/wsd/ PaySecureService.wsdl>
tModel / Interface: charge operation (document/literal, SOAP 1.1 over HTTP)
Description : Payment charging service for CampusEats orders

Task 6 :

Fault Mapping Note

Fault Mapping

When PaySecure returns a SOAP Fault, CampusEats maps it to its own internal error before sending any response to the student.

Mapping Example:

PaySecure Fault	CampusEats Internal Error	Message shown to Student
CARD_DECLINED	PAYMENT_DECLINED	Payment was declined. Please try another payment method.
INSUFFICIENT_FUNDS	PAYMENT_DECLINED	Payment was declined due to insufficient funds.
GATEWAY_UNAVAILABLE	PAYMENT_SERVICE_ERROR	Payment service is temporarily unavailable. Please try again later.
INVALID_CARD	INVALID_PAYMENT_DETAILS	Invalid payment details provided.